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Research Proposal
On

"A Hybrid Brain-Computer Interface for Speech and Exoskeleton Hand Movement in Nonverbal Autism"

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Research Topic:	**A Hybrid Brain-Computer Interface for Speech and Exoskeleton Hand Movement in Nonverbal Autism**

Introduction

Many autistic people reach adulthood without usable speech, and a quarter to a third remain minimally verbal [1]. For a large share of them, movement is also affected, most often as developmental coordination disorder [2]. Cerebral palsy appears in the same population far more often than it does in the general population [3]. Where the two conditions meet, a person can neither speak clearly nor use their hands reliably.

Augmentative and alternative communication was built for the first of those problems, but it assumes a steady touch. That assumption fails here, because a steady touch is exactly what the motor impairment removes. Brain-computer interfaces avoid the problem by reading the brain directly, yet the common designs create a new one. The P300 matrix speller, for example, flashes rows and columns of a letter grid on screen [4]. Its accuracy falls sharply once the user can no longer look straight at the letter they want [5]. Flashing light raises a second difficulty, since unusual sensory responses appear in the diagnostic criteria for autism [6]. Among those responses, dislike of flickering light is one of the most frequently reported [7]. Epilepsy is also more common here, and flashing between 15 and 25 Hz most often triggers seizures [8], [9]. None of this can simply be switched off, because the flicker is what creates the signal.

Research Problem

Taken together, these constraints leave the group without any single interface for speech and hand movement. A speech channel for these users would have to work without any flashing display at all. A powered hand would face a matching limit, since muscle and eye signals are not dependable here.

Research Question

> **Can one EEG recording support two decoding paradigms at the same time, a vibrotactile speller for speech and motor imagery for assisted grasp, in minimally verbal autistic users with co-occurring motor impairment?**

The speller reads a brief evoked response locked to each vibration, lasting well under one second. Motor imagery reads a slower change in the mu and beta rhythms across a window of seconds. The two signals differ in timing, in frequency band and in the electrodes that carry them. Vibration still reaches sensorimotor cortex and lowers the same rhythms that motor imagery is read from. The question is therefore how much accuracy the imagery channel loses while the speller runs.

Objectives

1) **Speller performance:** Measure online accuracy and information transfer rate for an eight-site vibrotactile speller across ten repetitions. The target is 70 percent mean accuracy against 12.5 percent chance, the usual minimum for useful communication [10].
2) **Imagery decoding:** Measure two-class imagery accuracy for left and right hand grasp, reported separately for each participant.
3) **Interference between paradigms:** Test whether concurrent vibration harms imagery decoding, since it disturbs the same sensorimotor rhythms [11].
4) **Assisted grasp:** Test whether the powered glove completes a set object-handling task, and relate the result to muscle tone.
5) **Sensory tolerability:** Test whether the calibrated vibration stays comfortable across a full session, and compare that against sensory profile scores.

Literature Review

Brouwer and van Erp established the basic result, showing that vibration at separate body sites works as oddball stimuli [12]. Building on that, van der Waal and colleagues produced a tactile speller as accurate as visual ones [13]. Severens and colleagues then compared the two in amyotrophic lateral

sclerosis, where the tactile route still worked [14]. Imagining a movement without making it lowers the mu and beta rhythms over the motor cortex [11]. That drop is called event-related desynchronisation, and Pfurtscheller and Neuper turned it into a control signal [15]. Since then, common spatial pattern filtering has become the standard way to separate left from right imagery [16]. The method is well established, although a small share of users produce no usable imagery signal [17].

Joining two methods is established: Pfurtscheller and colleagues defined the hybrid BCI as two signals from one recording [18]. Allison and colleagues then built a working hybrid of imagined movement and visual attention in healthy users [19]. On the hardware side, Soekadar and colleagues drove a hand exoskeleton from EEG and eye movement signals [20]. Imagery ability falls as motor impairment worsens, so usable imagery cannot be taken for granted here [21]. Autistic adults also feel vibration at lower levels than comparison groups in controlled laboratory tests [22].

Research Gap

Tactile BCIs give speech but not movement, while imagery BCIs give movement without a non-visual speller. Powered hand devices, for their part, rely on muscle or eye signals this group cannot produce reliably. No published system offers speech and grasp together to minimally verbal autistic users with motor impairment.

Methodology

1) **Design:** Bench testing will come first, and Phase 1 will then test both methods, separately and together, in thirty-six neurotypical adults.
2) **Participants:** Phase 2 will recruit thirty-six autistic participants with developmental coordination disorder or cerebral palsy and passively movable fingers. Muscle tone will be scored on the Modified Ashworth Scale, since tone sets the required force [23]. Because the glove covers both hands, the vibration units will sit on the forearm, upper arm and shoulder.
3) **Vocabulary:** Site 0 will always mean Stop, so the user can cancel an action with a single selection. The first selection will pick one of six message groups or grasp, and each group will offer seven items.
4) **Hardware:** Recording will use a sixteen-channel amplifier at 125 Hz, with Fz, Cz, CPz, Pz and POz for the speller. The imagery signal will come from C3, C4, CP3 and CP4, which sit over the motor cortex. Vibration will come from eight linear resonant actuators, driven by DRV2605L chips rather than simple rotating motors. Rotating motors were rejected because they start too slowly and blur the moment each vibration begins. Each burst will last 100 ms, a new one will start every 250 ms, and the order will be random. An optocoupler will mark every onset with a trigger pulse, keeping the two grounds apart, as Figure 1 shows.
5) **Exoskeleton:** A cable driven by a geared motor will close the hand, and an encoder will track its position. The motor will be sized against the resting muscle tone measured earlier, not against an unresisting finger. A torque sensor in line with each drive will report the force actually applied to the finger. Force will be limited twice over, by a current limit and a mechanical slip clutch, independently of software. A lever release will open the hand with the power off, and Figure 2 shows the whole drive.
6) **Decoding:** The system will not read thoughts, and will only detect which vibration the user is attending to. The speller path will filter between 0.1 and 12 Hz, then cut the signal into epochs. Those epochs will be classified with shrinkage-regularised linear discriminant analysis, which stays stable on few training examples [24]. The imagery path will work separately, filtering between 8 and 30 Hz before common spatial pattern filtering [16].
7) **Calibration:** Each site will be calibrated per user, starting with a two-down, one-up staircase for the detection threshold [25]. An ascending method of limits will then find the level at which it stops feeling comfortable.

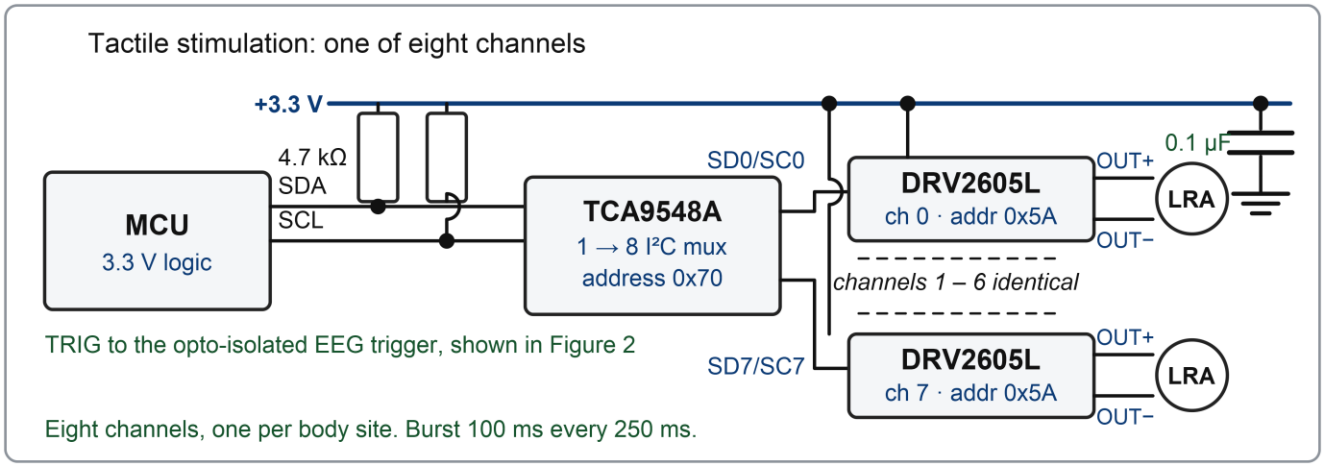


Figure 1: One of eight tactile stimulation channels. The DRV2605L has a fixed address, so a TCA9548A multiplexer reaches all eight, and each driver has a 0.1 microfarad decoupling capacitor.

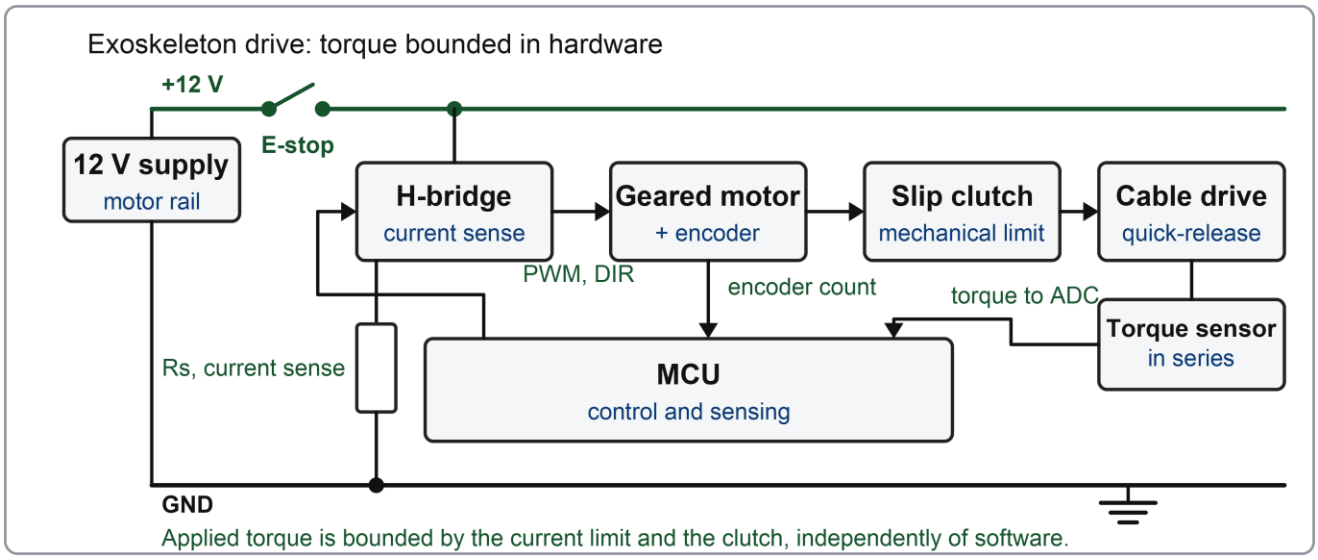


Figure 2: The exoskeleton drive. The current-sense resistor, the hardware current limit and the slip clutch limit the applied force without any software involved.

8) **Safety:** The glove will move a hand that may stiffen on its own, worn by someone who cannot report pain. A physiotherapist will measure how far each joint moves passively, and those figures will become the mechanical end-stops.
9) **Measures and ethics:** The main outcomes will be online speller accuracy and two-class imagery accuracy under normal operating conditions. Alongside these, the study will record information transfer rate, the accuracy change during vibration, and grasp success against Ashworth grade. Consent will follow each participant's usual way of communicating, with a supporter present, and distress will end the session.

Novelty and Expected Outcomes

The tactile oddball task [12], [13], common spatial pattern decoding [16] and the hybrid concept [18] all exist.

- **A hybrid interface for autism with co-occurring motor impairment:** Tactile BCI work has so far served locked-in and motor-impaired people who are not autistic. Autism BCI work trains attention and returns no output channel, so neither line reaches this overlap.
- **A tactile rather than visual second paradigm:** Existing hybrids pair motor imagery with a visual task, which brings the flicker straight back [19]. Replacing it with a tactile speller will remove flicker, gaze dependence and seizure risk in one step.
- **Interference treated as a measured question:** Existing hybrids assume their two channels are independent, and none reports the cost of that assumption [19].
- **An exoskeleton built to assist rather than to demonstrate:** Force will be set from the muscle tone measured earlier and limited in hardware, not software. That choice follows from the user, who may stiffen without meaning to and cannot describe pain.

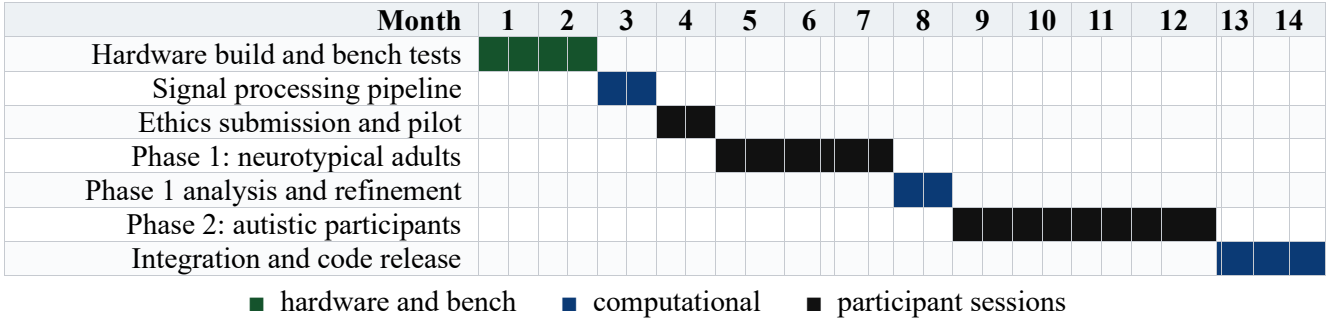
The outputs will be an open hardware and software design, per-participant accuracy for both methods, and an interference result.

Limitations

The samples will be small, so the study will show whether the system works at all rather than how well. Imagery may also be weaker in this group, and some participants will miss the target in objective 2 [17]. Comfort will be judged partly by observation, since minimally verbal participants cannot always describe discomfort themselves.

Feasibility and Timeline

The amplifier, actuators, drivers and multiplexer will come to about fifteen hundred US dollars at current prices. Beyond the hardware, the team will need EEG signal processing and embedded skills, plus a qualified physiotherapist. Recruitment will run through neurology, rehabilitation and autistic-led groups, contacted before the ethics application goes in. The schedule below runs for fourteen months, and no participant session will happen before ethics approval in month four.



Conclusion

This proposal replaces the screen with vibration and adds motor imagery to drive an assistive robotic hand. One recording then carries both a voice and a grasp, although neither part is new on its own. What no current system offers is both together, to users with neither steady gaze nor reliable movement. The wider value lies in what the channel carries, since a user who can report pain is safer.

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